1. Software Overview

The software controls the full functionality of the WiFi-enabled toy car. It enables communication between the mobile app and the ESP32, allowing the user to control movement, LED indicators, and the reverse assist system with safety features.

- The ESP32 (MicroPython) receives commands and controls motors, LEDs, ultrasonic sensor, and buzzer.
- The mobile app (MIT App Inventor) acts as a remote control, sending commands over WiFi.
- The system also includes an automatic safety stop, where the car stops when an obstacle is detected very close during reverse.

Platforms Used:

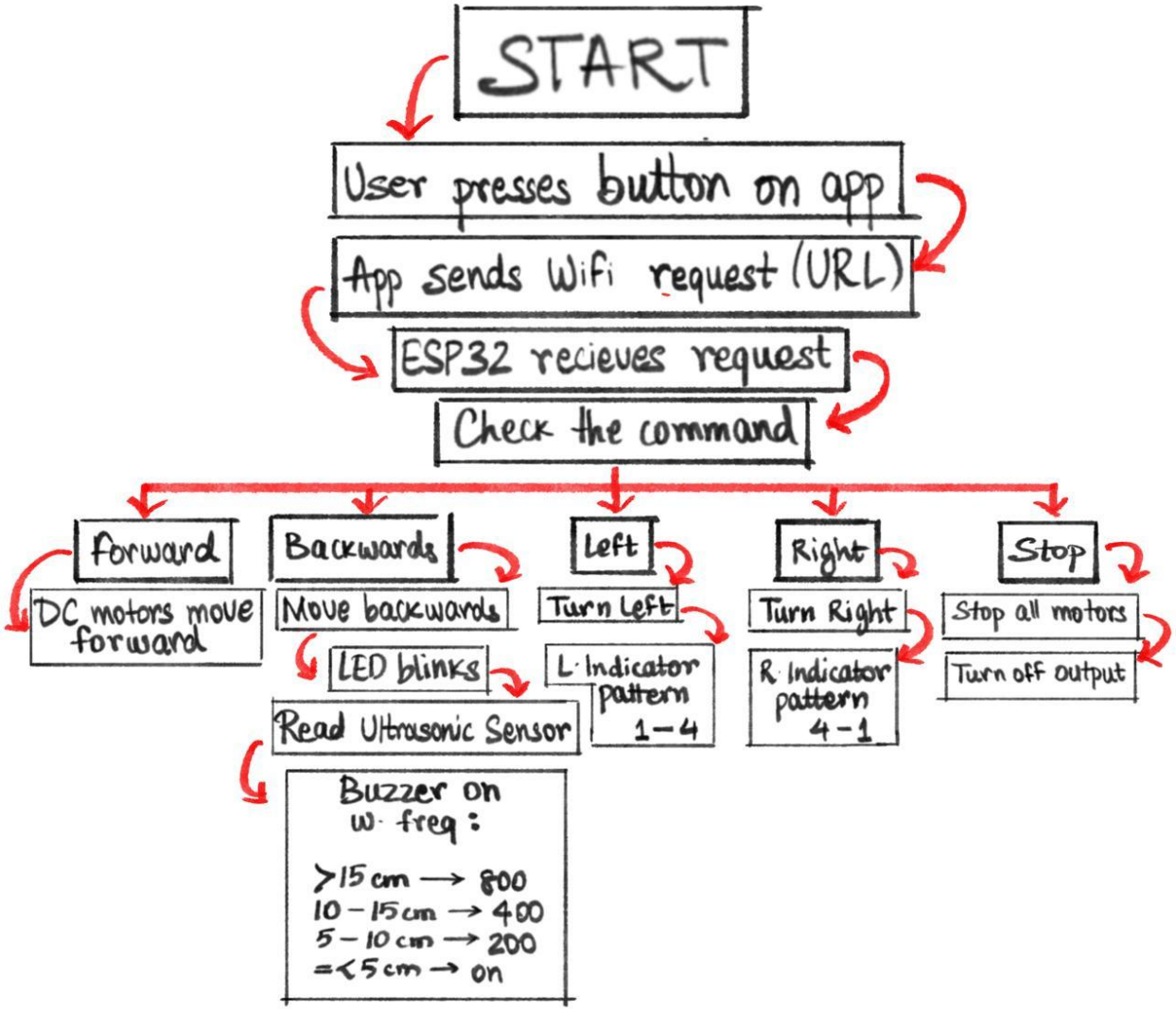
- MicroPython (ESP32 programming in Thonny)
- MIT App Inventor (mobile app interface)
- Cirkit Designer (circuit design and planning)

2. System Logic

1. User presses a button on the mobile app
2. App sends a command via WiFi (HTTP request)
3. ESP32 receives the command
4. Based on the command:
    - Motors move (forward / backward / left / right / stop)
    - LEDs activate (turn indicators or reverse blinking)

5. If the car is reversing:
    - Ultrasonic sensor continuously measures distance
    - Buzzer changes frequency based on distance:
        - Far → slow beeps
        - Near → faster beeps
        - Very close (< 5 cm) → continuous beep
6. Safety Condition:
    - If distance is less than 5 cm → car automatically stops
    - Backward motion is disabled to prevent collision
7. System continuously updates all outputs in real time based on user input and sensor data

3. Flowchart


START
User presses button on app
App sends Wifi request (URL)
ESP32 recieves request
Check the command
Forward
DC motors move forward
Backwards
Move backwards
LED blinks
Read Ultrasonic Sensor
Buzzer on w. freq:
>15 cm → 800
10 – 15 cm → 400
5 – 10 cm → 200
=<5 cm → on
Left
Turn Left
L. Indicator pattern 1 – 4
Right
Turn Right
R. Indicator pattern 4 – 1
Stop
Stop all motors
Turn off output

4. App Flow (If MIT App Inventor)

Screen Flow:

- Single screen interface
- Contains 5 main buttons: Forward, Backward, Left, Right, Stop

Button Actions:

Button	Action
Forward	**Sends /F → Car moves forward**
Backward	**Sends /B → Car reverses + LEDs blink + buzzer activates**
Left	**Sends /L → Car turns left + left LEDs blink**
Right	**Sends /R → Car turns right + right LEDs blink**
Stop	**Sends /S → Car stops**

Data Flow:

1. Button click → sets URL (e.g., http://192.168.4.1/F)
2. App sends request using Web component (Web1.Get)
3. ESP32 receives request via WiFi server
4. ESP32 executes the corresponding function
5. Physical components respond (motors, LEDs, buzzer)

Additional Logic (Reverse Safety Feature):

- When /B (Backward) is pressed:
    - LEDs blink at 0.5-second intervals

    - Buzzer starts reverse warning sound
    - Ultrasonic sensor continuously checks distance
- Based on distance:
    - Far → slow beeps
    - Near → faster beeps
    - < 5 cm → continuous beep + car automatically stops
- After stopping:
    - Backward motion is disabled until a new command is given

5.Final Code Used:

https://github.com/naishark06/ODT_Group-Work_Repo/blob/main/Final_Final_Car_Code.py